Countably Saturated Banach Spaces Have Countable Capture

Run `fa_banach_001`, lane 19

June 17, 2026

Source Problem

The source paper is Jinghao Huang, Karimbergen Kudaybergenov, and Rui Liu, *The ball-covering property of von Neumann algebras and noncommutative symmetric spaces*, arXiv:2311.14257. After proving the commutative $C(K)$ criterion and the von Neumann algebra criterion, it asks for a noncommutative C^*-algebra version:

> $\{U_\alpha\}_{\alpha\in A}$ of K is called a π-basis (weak basis) for K if every nonempty open subset of K contains some U_α in $\mathcal{U}$ [37, Page 15].
>
> In the setting of commutative C^*-algebra, [25, Theorem 2.1] can be reformulated as follows. Note that [25, Theorem 2.1] only deals with real continuous functions. The complex case is a direct consequence of the real case.
>
> **Corollary 3.9.** *Let $\mathcal{A}$ be a unital commutative C^*-algebra. Then the followings are equivalent:*
>
> (1) *$\mathcal{A}$ has the BCP;*
> (2) *the pure state space $\mathfrak{P}(\mathcal{A})$ has a countable π-basis;*
> (3) *$\mathcal{A}$ has the UBCP.*
>
> In particular, there exist commutative C^*-algebras containing no minimal projections but having the BCP. For example, the uniform norm closure of the linear span of Haar functions [24, p.150].
>
> We conclude this section by the following question.
>
> **Question 3.** *It will be interesting to find a non-commutative version of Corollary 3.9, that is, how to characterize those C^*-algebras $\mathcal{A}$ having BCP/UBCP.*

Definitions

Let X be a complex Banach space. We use the following elementary one-variable saturation principle. Say that X is *countably norm-saturated* if every countable family of closed norm conditions in one variable u, with parameters from X, that is finitely satisfiable in the unit ball of X, is realized in the unit ball of X.

This is weaker than full countable saturation in continuous logic. For C^*-algebras, it is also weaker than the usual model-theoretic countable saturation, and is covered by countable degree-one saturation for the specific linear norm conditions used below.

Statement

Theorem 1. *Let X be an infinite-dimensional countably norm-saturated complex Banach space. Then X has countable capture. Consequently X fails BCP.*

Corollary 1. *Every infinite-dimensional countably saturated C^*-algebra fails BCP. The same conclusion holds under any saturation hypothesis that realizes countable types consisting of the degree-one norm conditions in the proof.*

Idea of the Proof

For a finite set of centres $F \subset X$, choose norming functionals $f_c \in J(c)$. Since X is infinite dimensional, the finite intersection $\bigcap_{c \in F} \ker f_c$ contains a nonzero vector. A unit vector u in this intersection satisfies

$$\|c - \lambda u\| \geq |f_c(c - \lambda u)| = \|c\|$$

for every scalar λ. Thus finite pieces of the countable-capture requirements are always satisfiable.

For a countable centre set C, write the requirements using only rational complex scalars:

$$\|u\| = 1, \qquad \|c - \lambda u\| \geq \|c\| \quad (c \in C,\ \lambda \in \mathbb{Q} + i\mathbb{Q}).$$

Countable saturation realizes these simultaneously. Continuity in λ upgrades the inequalities from rational scalars to all complex scalars, giving $\mathrm{dist}(c, \mathbb{C}u) = \|c\|$ for every $c \in C$.

Proof

Proof of Theorem 1. Let $C = \{c_1, c_2, \ldots\} \subset X$ be countable. Consider the following countable type in the variable u:

$$\|u\| = 1,$$

and, for every $j \geq 1$ and every $\lambda \in \mathbb{Q} + i\mathbb{Q}$,

$$\|c_j - \lambda u\| \geq \|c_j\|.$$

Equivalently, the latter can be written as the closed norm condition

$$\max\{0, \|c_j\| - \|c_j - \lambda u\|\} = 0.$$

We first prove finite satisfiability. Fix finitely many centres $c_1, \ldots, c_N$. Discard the zero centres. For each remaining c_j, choose $f_j \in X^*$ such that

$$\|f_j\| = 1, \qquad f_j(c_j) = \|c_j\|.$$

The space

$$\bigcap_{j=1}^{N} \ker f_j$$

has finite codimension in the infinite-dimensional space X, and is therefore nonzero. Choose u in this intersection with $\|u\| = 1$. Then for every listed centre and every $\lambda \in \mathbb{C}$,

$$\|c_j - \lambda u\| \geq |f_j(c_j - \lambda u)| = \|c_j\|.$$

The zero centres satisfy the same inequality trivially. Hence every finite part of the type is satisfiable.

By countable norm-saturation, there is $u \in X$ realizing the whole type. Then $\|u\| = 1$, and for every j and every rational complex λ,

$$\|c_j - \lambda u\| \geq \|c_j\|.$$

For fixed j, the map $\lambda \mapsto \|c_j - \lambda u\|$ is continuous, and $\mathbb{Q} + i\mathbb{Q}$ is dense in $\mathbb{C}$. Therefore

$$\|c_j - \lambda u\| \geq \|c_j\| \qquad (\lambda \in \mathbb{C}).$$

Taking the infimum over λ gives

$$\operatorname{dist}(c_j, \mathbb{C}u) \geq \|c_j\|.$$

The reverse inequality follows by taking $\lambda = 0$. Hence

$$\operatorname{dist}(c_j, \mathbb{C}u) = \|c_j\| \qquad (j \geq 1).$$

By the countable-capture characterization from the earlier lane-19 packet, X has countable capture. The countable-capture obstruction then implies that X fails BCP. □

Proof of Corollary 1. A countably saturated C^*-algebra is countably saturated as a metric structure, so in particular it realizes the one-variable norm type used in Theorem 1. Apply the theorem to the underlying Banach space. □

Scope for the C*-Classification

This result adds a model-theoretic negative mechanism to the previous topological, tensorial, product, von Neumann, group, and corona mechanisms. It explains why ultrapower and reduced-power style C^*-algebras naturally fall on the negative side: the finite centre obstruction is automatic in infinite dimension, and saturation turns finite satisfiability into a single global escaping direction.

The theorem does not settle arbitrary simple nonseparable C^*-algebras of density continuum, because such algebras need not be countably saturated or even countably degree-one saturated.

Novelty and Search Bounds

The local run indexes were searched for saturation, countable capture, ultrapowers, reduced powers, and countably degree-one saturation. No prior lane-19 packet recorded this general saturation-to-capture implication. Web searches on June 17, 2026 found nearby saturation literature for C^*-algebras and coronas, including Farah–Hart on countable saturation of corona algebras, Eagle–Vignati on saturation and elementary equivalence of C^*-algebras, and Voiculescu on countable degree-one saturation.

Human Review Focus

Reviewers should check:

1. the finite satisfiability argument via finitely many norming functionals;
2. that the countable type uses only one-variable norm conditions;
3. the passage from rational scalars to all complex scalars;
4. the stated relationship between the custom saturation hypothesis and standard countable saturation/countable degree-one saturation.

References

[1] Jinghao Huang, Karimbergen Kudaybergenov, and Rui Liu, *The ball-covering property of von Neumann algebras and noncommutative symmetric spaces*, arXiv:2311.14257, 2023.

[2] Ilijas Farah and Bradd Hart, *Countable saturation of corona algebras*, arXiv:1112.3898, 2011.

[3] Christopher J. Eagle and Alessandro Vignati, *Saturation and elementary equivalence of C^*-algebras*, arXiv:1406.4875, 2014.

[4] Dan-Virgil Voiculescu, *Countable Degree-1 Saturation of Certain C*-Algebras Which Are Coronas of Banach Algebras*, arXiv:1310.4862, 2013.

[5] Run `fa_banach_001`, lane 19, *When the Countable-Capture BCP Obstruction Exists*, solution packet `2311.14257_countable_capture_characterization`, 2026.